

Trained

Large labelled
training set

train

Corpus-specialized
features

Standard

Cross-env.



Standard



Cross-env.



Training-free, prior work

Frozen patch features

Codebook or Segmentation CLS attention or

Prior-selected / aggregated features

Standard **X** Cross-env. —



Frozen patch features

Codebook Segmentation CLS attention

Standard



Cross-env.



Ours

Frozen patch features → Within-image → Across-image

Standard ✓

Cross-env. ✓

Within-image: Balanced, matchable patch evidence



Frozen patch features

Within-image	Across-image
Balanced, matchable patch evidence	

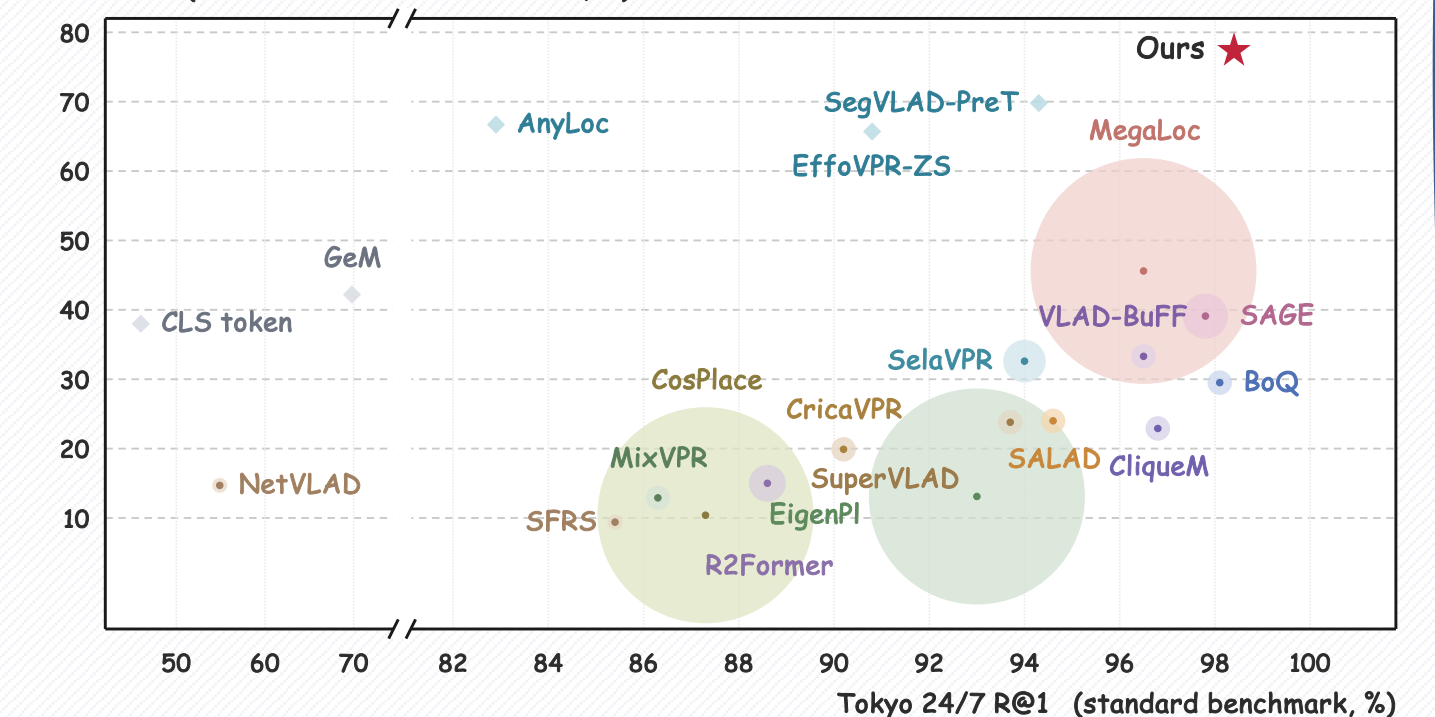
Standard



Cross-env.



VP-Air R@1 (cross-environment benchmark, %)



★ ours (training-free)

► training-free

- **trained**

● 1 million images

5 million images

Performance across 17 benchmarks

standard

cross-environment

Legend:

- Ours
- - - MegaLoc
- - - AnyLoc

Benchmarks (clockwise from top):

- Pitts250k
- Pitts30k
- Tokyo 24/7
- MSLS-val
- Nordland
- St Lucia
- Oxford
- AmsterTime
- 17 Places
- Baidu Mall
- Gardens Point
- VP-Air
- Nardo-Air
- Nardo-Air R
- Hawkins
- Laurel
- Mid-Atlantic

Performance Scale (Concentric Dashed Circles):

- 20
- 60
- 100

